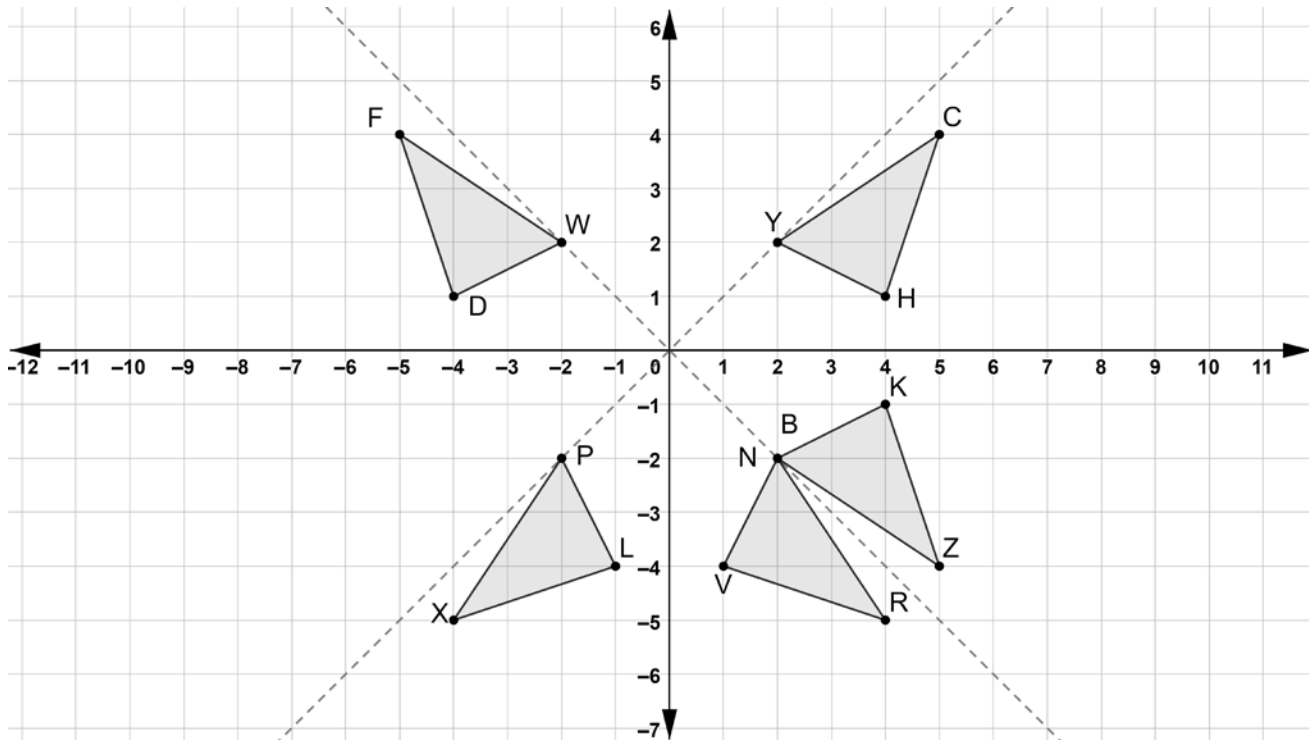


Geometry  
Unit 4  
Lesson 2 Practice

Name \_\_\_\_\_  
Date \_\_\_\_\_  
Period \_\_\_\_\_

Six triangles are shown on the coordinate grid.



1. Draw the reflection of  $\triangle PLX$  across the line  $y = x$ . Label the reflection as  $\triangle P'L'X'$ .
2. Draw the reflection of  $\triangle FWD$  across the line  $y = -x$ . Label the reflection as  $\triangle F'W'D'$ .
3. Select all of the triangles that are reflections of  $\triangle CHY$ .
  - ☐  $\triangle FDW$
  - ☐  $\triangle XLP$
  - ☐  $\triangle RVN$
  - ☐  $\triangle ZKB$
  - ☐  $\triangle P'L'X'$
  - ☐  $\triangle F'W'D'$

4. The algebraic description of the reflection of  $\triangle BKZ$ , the preimage, onto  $\triangle NVR$ , the image, is  $(x, y) \rightarrow (-y, -x)$ . Describe how the coordinates of the vertices of  $\triangle BKZ$ , the preimage, and the coordinates of  $\triangle NVR$  the image can be used to determine the line of reflection.
5. Which triangle is the image of reflecting  $\triangle FDW$  across the  $x$ -axis?
6. Which triangle is the image of reflecting  $\triangle CHY$  across the  $y$ -axis?
7. Describe how the coordinates of the vertices of  $\triangle F'W'D'$ , the preimage, and the coordinates of  $\triangle XPL$  the image can be used to determine the algebraic description of the reflection.
8. Describe how the coordinates of the vertices of  $\triangle ZKB$ , the preimage, and the coordinates of  $\triangle CHY$ , the image can be used to determine the algebraic description of the reflection.
9. Write the algebraic description for the reflection of  $\triangle NVR$ , the preimage, onto  $\triangle PLX$ , the image.
10. Write the algebraic description for the reflection of  $\triangle P'L'X'$ , the preimage, onto,  $\triangle BKZ$ , the image.